

Demonstrating Haptic Source-Effector: Full-Body Haptics via Non-Invasive Brain Stimulation

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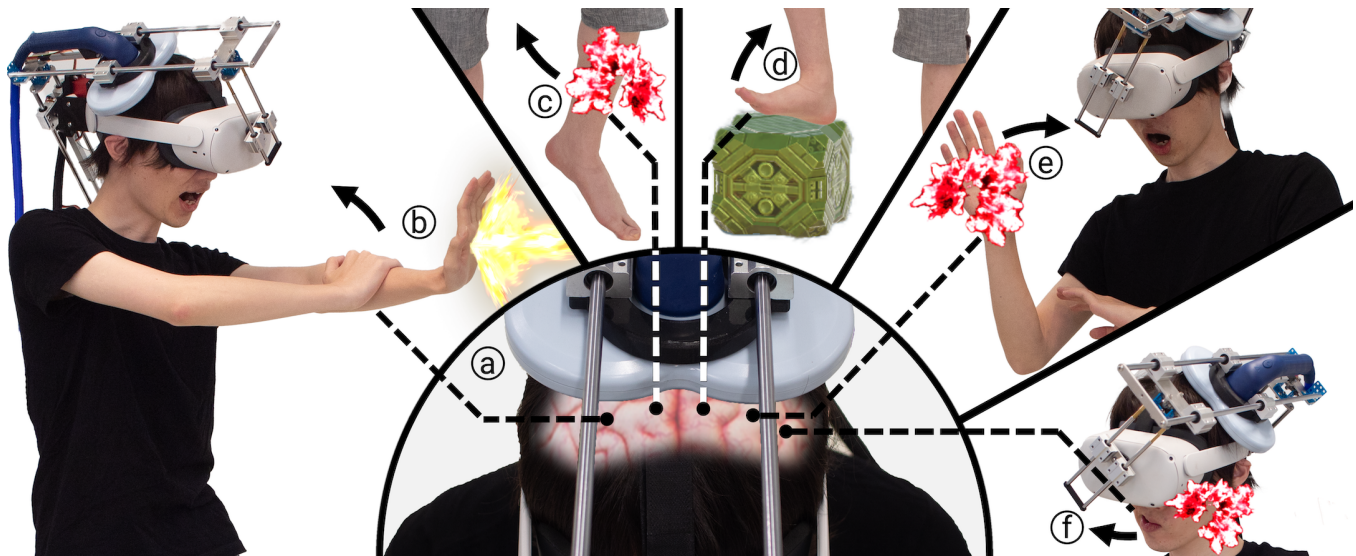


Figure 1: We propose a novel concept in which a single on-body actuator renders haptics to multiple body parts—even as distant as one’s foot or one’s hand—by stimulating the user’s brain (a). We implemented this by mechanically moving a coil across the user’s scalp. As the coil sits on specific regions of the user’s sensorimotor cortex it uses electromagnetic pulses to non-invasively & safely create haptic sensations, e.g., touch and/or forces. For instance, (b) recoil of throwing a projectile, (c) impact on the leg, (d) force of stomping on a box, (e), impact of a projectile on one’s hand, or (f) an explosion close to the jaw.

ABSTRACT

We propose a novel concept for haptics in which one centralized on-body actuator renders haptic effects on multiple body parts by stimulating the brain, i.e., the source of the nervous system—we call this a haptic source-effector, as opposed to the traditional wearables’ approach of attaching one actuator per body part (end-effectors). We implement our concept via transcranial-magnetic-stimulation (TMS)—a non-invasive technique from neuroscience/medicine in which electromagnetic pulses safely stimulate brain areas. Our approach renders ~15 touch/force-feedback sensations throughout the body (e.g., hands, arms, legs, feet, and jaw), all by stimulating

the user’s sensorimotor cortex with a single magnetic coil moved mechanically across the scalp.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; • **Hardware** → **Emerging interfaces**.

KEYWORDS

Transcranial Magnetic Stimulation, Electrical Muscle Stimulation, Haptics

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1 INTRODUCTION

The overwhelming majority of haptic devices are engineered using the same principle, i.e., an actuator is mounted atop the part of the skin that is meant to be stimulated, e.g., finger-worn haptic devices deliver tactile cues to the fingerpad [11], foot-worn devices deliver tactile cues to the soles [12], head-worn devices deliver tactile cues to the face [1], and so forth—this approach can also be referred to as haptic *end-effector* [4], a term borrowed from robotics literature, where it denotes how the actuator is attached at the very *endpoint* of a robotic extremity. Similarly, in this context, a haptic end-effector denotes how the actuator is attached at the very *endpoint* of the user’s nervous system, e.g., a vibration motor attached atop the tactile receptors at one’s fingertips [14]. Users feel haptic sensations delivered by end-effectors via the following process: (1) mechanical stimuli are translated to electrical impulses (by mechanoreceptors at the endpoint), (2) electrical signals propagate via the peripheral nervous system (via neurons from limbs to the spinal cord), and, finally, (3) signals are processed in the *source* location where all the neurons flow (i.e., the brain, in particular, the somatosensory cortex).

This end-effector approach has been successful in that devices can deliver precise temporal and spatial cues. However, few examples of these devices lend themselves well to *full-body* haptic experiences, which reveals a research question to which our field has not found new answers: how might we *scale up* haptic devices given that the current approach requires *many* devices to render sensations on *many* locations of the user’s body?

Thus, to propose a new approach, we take the inverse conceptual turn. Instead of placing haptic end-effectors on the endpoints of the user’s nervous system, we propose delivering full-body haptics via a single haptic device that directly stimulates the user’s brain areas responsible for creating haptic sensations—we call this a *source-effector*.

To create the first instantiation of this concept, we turned to transcranial magnetic stimulation (TMS)—a technique from medicine that allows for safe & non-invasive brain stimulation. To translate the TMS technique into readily usable knowledge for human-computer interaction, we engineered a haptic device that sits atop the user’s head and is able to position the TMS coil at strategic locations to generate haptic sensations (i.e., either touch or touch & force) in nine locations of the users’ body: hands, arms, legs, feet, and jaw, as shown in Figure 1—all by stimulating the user’s sensorimotor cortex with a single magnetic-coil moved mechanically across the scalp. The details regarding user evaluations as well as the roadmap to extend interactive opportunities of TMS can be found in [13].

2 IMPLEMENTATION

To help readers replicate our design, we now provide the necessary technical details (hardware & software)¹.

Hardware Implementation. We engineered the actuation system shown in Figure 2, capable of moving a TMS coil (Magstim D70² butterfly coil) around the user’s head, which is connected to a medical-grade magnetic stimulator (Magstim Super Rapid²).

Robotic platform. The key component in our hardware implementation is a robotic gantry that mechanically moves the TMS coil across key areas of the user’s scalp. Our design is inspired by a traditional X-Y-Z gantry system commonly found in CNC machines [7], but with key modifications that allow it to: (1) most importantly, conform to the curvature of the scalp around the pitch axis (i.e., front/back), which is estimated to be ~18 degrees for the sensorimotor cortex area, based on our measurement from a standard head shape dataset [6]; (2) accommodate different heads, including different curvatures and sizes; (3) actuate with sufficient force to move a medical-grade TMS coil (~1 kg); (4) actuate with steps smaller than 8.5 mm, as characterized by a user study in [13]; and, finally, (5) provide a structure that can be either directly mounted to a VR headset or suspended from the ceiling (as in [8]). To satisfy these requirements we engineered the actuation system shown in Figure 2. Further details on our device’s actuators and structural elements can be found in [13] alongside technical evaluations on the accuracy of controlling the coil position with our device.

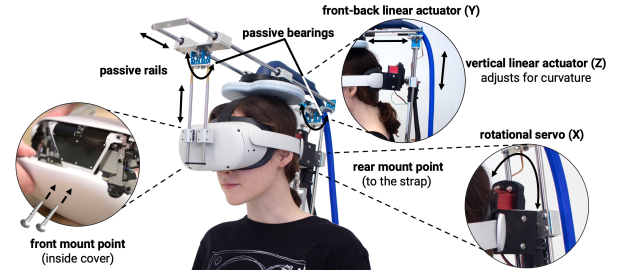


Figure 2: Overview of our mechanical actuator to interactively displace the TMS coil over key areas of the user’s brain.

Stimulation parameters. We use three consecutive TMS pulses with 50 ms intervals as our stimulation. Regarding repetitive TMS stimulation at this 50 ms interval, Rossi et al. suggested fewer than eight pulses [10], a guideline we strictly followed. Furthermore, we hardcoded a condition that TMS stimulation can only be activated after five seconds have elapsed prior to the preceding stimulation. This was also incorporated into our game design, by adjusting aspects such as the timing of the enemy’s attack or charge-up duration for the user’s projectile launcher—all in adherence to Rossi et al.’s safety guidelines [10].

Software Implementation. To control our medical-grade TMS stimulator, we implemented a custom middleware via MagPy Python Toolbox [9], which routed serial communication to the stimulator and received Open Sound Control (OSC) commands. Our main applications for both studies ran on Unity3D, which dispatched OSC commands to our custom middleware for TMS, and serial commands to a SAMD21-XIAO microcontroller connected to the motors and encoders. For tracking the hands and feet, we used HTC VIVE 3.0 Trackers attached to the user’s extremities, and integrated their tracking information into Quest 2’s headset tracking using Open VR Space Calibrator [5].

Safety. As one of the very first explorations of TMS for haptics, we believe that discussing safety guidelines is crucial for wider research. The safety protocol for TMS largely overlaps with EMS’ [3], as both apply electrical currents (e.g., they both require per-user calibration, etc). Moreover, due to its use of an electromagnetic

¹<https://lab.plopes.org/#source-effector>

field, TMS also shares some of the safety guidelines of fMRI [2], i.e., remove any ferromagnetic materials from the stimulation range and maintain some distance from electronic devices to prevent unintended interactions. For runtime safety, we implemented an emergency stop function in our Unity application, which immediately shuts off both the TMS stimulation (hardware switch) and motor power (hardware switch).

3 PARTICIPANT EXPERIENCE

At UIST 2024 Demo, we will demonstrate the proposed haptic source-effector. A participant first consent to experience our system via ACM demo waiver. Then, they proceed with a safe TMS calibration, where a demonstrator finds the coil positions and intensities that induce involuntary movements of the hands, forearms, feet, and lower legs via TMS pulses. After that, they try the VR game illustrated in Figure 3. Participants embodied the avatar of a cyborg trying to escape a robotics factory that has malfunctioned, as depicted in Figure 3 (a). However, when they find the escape route blocked by malfunctioning robots that fire at them, —as depicted in Figure 3 (b)— the VR experience commands our haptic device to render *tactile sensation on the affected area* (e.g., the left hand in this case, but both hands and feet are possible). To advance, participants can counteract by charging up their plasma-hand and firing plasma-projectiles to deactivate the robots—as depicted in Figure 3 (c)—when they open the palm of their hands in a firing gesture, the VR experience detects this gesture and prompts our haptic device to render *force-feedback and tactile sensations on the firing hand* (e.g., the right hand in the case, but both hands can fire plasma shots). After this, the user continues to counteract any robots that appear, which can fire shots against any of the user’s VR limbs (i.e., the hands or feet). Figure 3 (d) shows the user being shot in their right foot—just before this happens, the VR prompts our haptic device to render *tactile sensation on the right foot*. After a while, the user’s plasma-hand stops working, and they need to recharge the energy. They locate a crate on the floor and stomp it with their feet to release its charging energy—as depicted in Figure 3 (e)—just before the stomping releases the energy, the VR commands our haptic device to render *force-feedback and tactile sensation on the left leg*. The user keeps fighting until they eventually find a button that opens the exit door—as depicted in Figure 3 (f)—as they are about to press it, the VR requests that our haptic device render *tactile sensation on the right hand*. The user has escaped the factory. Throughout the calibration and demonstration, the participant sits on our custom chair physically grounded to our device, which significantly reduces the device weight applied to the participant’s head, preventing the fatigue. Considering the limited time for demonstration, if necessary, the attendee can instead experience an application that requires only two haptic feedback locations (left foot for getting hit by a projectile and right hand for shooting a projectile), which substantially shortens calibration time.

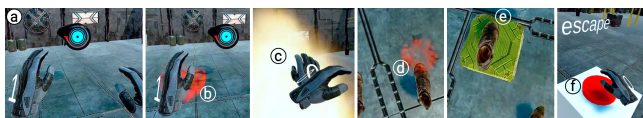


Figure 3: Full-body interactions in our demo, including: force-feedback and tactile sensations on both hands and both feet.

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